

Timothy Ojebiyi

London, United Kingdom | +44 7873 264900 | tobiojebiyi@gmail.com | [LinkedIn](#) | [GitHub](#) | [Website](#)

Professional Summary

Computer Science graduate (2:1, University of the West of England, Bristol) specialising in AI/ML and full-stack engineering. Solo founder of an AI-powered compliance SaaS, shipping production systems end-to-end from database design to deployment. Builds agentic AI systems with production-grade safety guardrails, and applies the same statistically rigorous, evidence-driven approach across machine learning, forecasting and systems engineering projects.

Education

BSc (Hons) Computer Science | University of the West of England, Bristol 09/2022 – 11/2025
Grade: 2:1 (Honours)

Experience

- Founder | RootLayer Ltd** 05/2026 – Present
 - Built and shipped an AI-powered web compliance SaaS solo, covering WCAG 2.2 AA accessibility and GDPR/EAA privacy auditing: a hybrid engine (an LLM agent driving a real browser through 12 audit tools on a deterministic axe-core baseline, plus a four-phase deterministic consent/tracker scanner), full product infrastructure (async job processing, multi-tenancy, white-label reporting, SSRF protection and rate limiting), and commercial outreach run in parallel.
- Full-Stack Engineer Intern | CarbonTrac** 07/2025 – 09/2025
 - Developed a PostgreSQL database with a 10-table star schema storing product emissions, supplier and shopping-history data.
 - Built an ML model analysing 46 nutritional parameters and a recommendation engine that detects nutrient gaps from shopping history.
- UX & Accessibility Intern | Nomensa** 07/2024 – 08/2024
 - Led UX research for an AI-powered healthcare product, running stakeholder workshops and user interviews, and developed personas and journey maps that identified 4 critical pain points in the patient experience.

Projects

- ColonoScan: Colon Cancer Detection Platform** · Final-year dissertation
Built a containerised medical AI platform combining histopathology and colonoscopy video analysis for colon cancer detection: a Next.js frontend, Django REST backend, and FastAPI/PyTorch microservices. Paired an EfficientNet-B4 UNet++ model for real-time polyp segmentation with explainable AI (Grad-CAM, saliency maps).
Technologies: Next.js, React, Django REST Framework, FastAPI, PyTorch, Celery, Redis, OpenCV, PostgreSQL, Docker
- Olimpia: AI Chessathon Engine** · Optiver-sponsored competition
Joined the Optiver-sponsored AI Chessathon 2026 on Day 3, climbing from #241 to #82 of 485+ teams in under 48 hours. Compiled the search with Numba (bitboards, magic attack tables, LLVM bit-scan intrinsics), lifting throughput from 13,000 to ~2 million nodes/second. Shipped only on statistically significant results at the platform's real-time control, including a time-management fix worth +61 Elo (95% CI [+35, +88], 513 games).
Technologies: Python, Numba (JIT), bitboards, alpha-beta/PVS search, transposition tables, logistic regression
- World Cup 2026 Forecasting Model**
Built a probabilistic forecasting model for the 2026 FIFA World Cup: an Elo-difference-parameterised Dixon–Coles model fit on ~49,000 international matches since 1872, wrapped in a Monte Carlo bracket simulation. Validated with a leak-free walk-forward backtest against a de-vigged bookmaker baseline, correctly ranking the eventual four semi-finalists as its top four by champion probability, and falsified a published RPS claim from a 2026 paper.
Technologies: Python, Dixon–Coles bivariate Poisson, World Football Elo, LightGBM, Monte Carlo simulation, walk-forward validation
- Credit Card Fraud Detection** · End-to-end ML analysis
Tackled credit card fraud as an extreme class-imbalance problem (0.172% of transactions fraudulent), comparing four model families and four imbalance-handling strategies, evaluated on business-relevant metrics. Best model, XGBoost with class weighting, caught 83.7% of fraud at a 0.04% false-alarm rate, worth \$9,811 net benefit per ~57,000 transactions.
Technologies: Python, scikit-learn, XGBoost, imbalanced-learn, pandas, NumPy

Skills & Technical Proficiency

Languages: Python, TypeScript, JavaScript, C++, SQL
Frontend: React, Next.js, Tailwind CSS, Responsive Design
Backend & Data: FastAPI, Django, Node.js/Express, REST APIs, PostgreSQL, Redis, Celery
ML & AI: PyTorch, scikit-learn, LightGBM, CNNs, LLM orchestration, pandas, NumPy
Tools: Docker, Git, AWS, CI/CD, Figma